fn make_report_noisy_top_k

Michael Shoemate

February 24, 2025

Proves soundness of `make_report_noisy_top_k` in `mod.rs` at commit f5bb719 (outdated[1]). `make_report_noisy_top_k` returns a Measurement that noisily selects the index of the greatest score from a vector of input scores. This released index can be later be used to index into a public candidate set (postprocessing).

1 Background

This mechanism fulfills the same purpose as the exponential mechanism, where the release is the best candidate's index from a finite set. The naive implementation of the exponential mechanism samples an index k from $[m] = 1, \ldots, m$, where m is the number of candidates, with probability p_i assigned to each candidate's index i as a function of their score s_i. The output is drawn via inverse transform sampling by outputting the smallest index k for which the cumulative probability is greater than some $u \sim \text{Uniform}(0, 1)$.

$$\mathcal{M}_{\text{naive}}([s_1, \ldots, s_m]) = \min\{k : \sum_{i=1}^{k} p_i \geq u\} \tag{1}$$

The probability of index k being selected is the normalization of its likelihood $\exp(s_k/\tau)$. As a candidate's score s_k increases, the candidate becomes exponentially more likely to be selected:

$$p_k = \frac{\exp(s_k/\tau)}{\sum_{i=1}^{m} \exp(s_i/\tau)} \tag{2}$$

This equation introduces a new temperature parameter, τ, which calibrates how distinguishable scores are from each other. As temperature increases, the categorical output distribution tends towards higher entropy/uniformity and becomes more privacy preserving. As temperature decreases, the categorical distribution tends towards a one-hot vector (where each candidate has zero probability, except for the candidate with the maximum score, which has probability one), becoming less private. Temperature is related to the privacy loss parameter (`d_out`) and sensitivity of the scoring function (Δ) as follows:

$$\tau = \Delta/\texttt{d_out} \tag{3}$$

A precise definition of Δ will come later and is captured by the metrics we use on the score vector $s = [s_1, \ldots, s_m]$. When `d_out` increases, temperature decreases, and candidates become more distinguishable from each other. We also divide scores by their global sensitivity to normalize the sensitivity to one. In the differential privacy literature for the exponential mechanism, the sensitivity is often multiplied by two. In OpenDP's `make_report_noisy_top_k` this factor is bundled into the Δ term, which is expressed in terms of a metric that captures monotonicity.

[1] See new changes with `git diff f5bb719..082a7224 rust/src/measurements/report_noisy_top_k/mod.rs`

1.1 Sampling Vulnerabilities

In practice, computing $\exp(s_i/\tau)$ is prone to zero underflow (where a non-zero quantity rounds down to zero) and overflow (where a large finite quantity is replaced with infinity) due to finite/limited data representation. Specifically, a scaled score s_i/τ of just -709 underflows to zero and $+710$ overflows to infinity when stored in a 64-bit float.

A simple improvement is to shift the scores by subtracting the greatest score from all scores. In idealized arithmetic, the resulting probabilities are not affected by shifts in the underlying scores. On finite data types, this shift prevents a catastrophic overflow, but makes underflow more likely, causing tail values of the distribution to round to zero. The inverse transform sampling step is also subject to accumulated rounding errors from the arithmetic and sum, which influence the likelihood of being chosen.

These potential vulnerabilities can be addressed via the Gumbel-max trick. The naive mechanism $\mathcal{M}_{\text{naive}}$ implemented with infinite-precision arithmetic is equivalent in distribution to the following mechanism:

$$\mathcal{M}([s_1, \ldots, s_m]) = \text{argmax}_i g_i, \tag{4}$$

where each $g_i \sim \text{Gumbel}(\mu = s_i, \beta = \tau)$.

1.2 Noise Distribution

`make_report_noisy_top_k` can also be configured to satisfy either `MaxDivergence` or `RangeDivergence`. Gumbel noise is used when `output_measure` is `RangeDivergence`, and exponential noise is used when `output_measure` is `MaxDivergence`. These choices of noise distributions minimize the necessary noise variance for their respective privacy measures.

2 Hoare Triple

Precondition

Compiler-verified

- `MO` is a type with trait `SelectionMeasure`
- `TIA` (atomic input type) is a type with trait `Number`

Caller-verified

None

Pseudocode

```
def make_report_noisy_top_k(
    input_domain: VectorDomain[AtomDomain[TIA]],
    input_metric: RangeDistance[TIA],
    privacy_measure: MO,
    k: int,
    scale: f64,
    optimize: Literal["max", "min"],
) -> Measurement:
    if input_domain.element_domain.nullable:  #
        raise ValueError("input domain must be non-nullable")

    if input_domain.size is not None:
        if k > input_domain.size:
            raise ValueError("k must not exceed the number of candidates")

```

```
    if k > 1 and not MO.ONE_SHOT:  #
        raise ValueError("privacy measure must support one-shot")

    if scale.is_sign_negative():  #
        raise ValueError("scale must be non-negative")

    f_scale = FBig.try_from(scale)  #

    if f_scale.is_zero():
        # ZERO SCALE
        function = Function.new_fallible(function_report_top_k(k, optimize))

    else:
        # NON-ZERO SCALE
        function = Function.new_fallible(
            function_report_noisy_top_k(k, f_scale, optimize)
        )

    def privacy_map(d_in: TIA):  #
        # convert to range distance
        # will multiply by 2 if not monotonic
        d_in = input_metric.range_distance(d_in) #

        d_in = f64.inf_cast(d_in)  #

        #
        return privacy_measure.privacy_map(d_in, scale, k)

    return Measurement.new(
        input_domain=input_domain,
        function=function,
        input_metric=input_metric,
        output_measure=privacy_measure,
        privacy_map=privacy_map,
    )
```

Postcondition

Theorem 2.1. For every setting of the input parameters `input_domain, input_metric, output_measure, k, scale, optimize, MO, TIA` to `make_report_noisy_top_k` such that the given preconditions hold, `make_report_noisy_t` raises an exception (at compile time or run time) or returns a valid measurement. A valid measurement has the following property:

1. (Privacy guarantee). For every pair of elements x, x' in `input_domain` and for every pair (`d_in`, `d_out`), where `d_in` has the associated type for `input_metric` and `d_out` has the associated type for `output_measure`, if x, x' are `d_in`-close under `input_metric`, `privacy_map`(`d_in`) does not raise an exception, and `privacy_map`(`d_in`) $\leq$ `d_out`, then `function`(x), `function`(x') are `d_out`-close under `output_measure`.

3 Proof

3.1 Data-independent runtime errors.

There are two sources of runtime errors in the function:

`greater_than`, which can in turn only occur due to lack of system entropy. This kind of failure is generally considered data-independent, where a lack of system entropy would occur regardless of the choice of input datasets. However, failure due to lack of entropy can be data-dependent in this case.

An input score vector with all the same scores is expected to require more draws from the random number generator, as the candidates will be very competitive, as compared to a score vector with widely different

scores. This technically results in input datasets with more homogeneity being more likely to exhaust entropy and raise an error, violating the data-independent runtime error requirement. This is an unlikely exploit in practice, due to the difficulty of exhausting the RNG's entropy.

The data-independent runtime error requirement is otherwise satisfied.

3.2 Privacy Guarantee

Definition 3.1. $\mathcal{M}_{\mathrm{RV}}(x)$ is a noninteractive mechanism that, when passed a vector of non-null scores, returns the indices of the top k noisy processed scores z_i, where each $z_i \sim \mathrm{RV}(\mathrm{shift} = y_i, \mathrm{scale} = \texttt{scale})$, and each $y_i = -x_i$ if `optimize` is `min`, else $y_i = x_i$.

Theorem 3.2. `function`implements $\mathcal{M}$.

`function`will return a data-dependent error if entropy is exhausted. Otherwise the returned function will only error if there are no non-null scores.

Proof of Theorem 3.2. By line 19 the scale is non-negative.

If scale is zero, then since the preconditions for `function_report_top_k` are met, then by its postcondition, since scale is zero, `function`is equivalent to $\mathcal{M}_{\mathrm{RV}}(x)$ as defined in 3.1. This is because RV is from a location-scale family of distributions, and the scale parameter is zero.

If scale is positive, then since the preconditions for `function_report_noisy_top_k` are met, then by its postcondition, since scale is positive, `function`is equivalent to $\mathcal{M}_{\mathrm{RV}}(x)$.

Therefore, `function`is equivalent to $\mathcal{M}_{\mathrm{RV}}(x)$. □

Proof for 2.1 Privacy Guarantee. The privacy map is defined on line 34. `d_in`denotes the greatest distance between adjacent score vectors in terms of the `LInfDistance` metric. By the postcondition of `LInfDistance.range_distance`, `d_in` is converted to the range distance, and on line 39 the distance type is conservatively casted to float.

By Theorem 3.2, `function`implements $\mathcal{M}_{\mathrm{RV}}$, where RV denotes the associated noise distribution of `output_measure` as defined in `SelectionMeasure`. Since `d_in`denotes the range distance, and `function`implements $\mathcal{M}_{\mathrm{RV}}$, then the preconditions for `SelectionMeasure.privacy_map` on line 41 are met. The postcondition on the value returned is then consistent with the privacy guarantee of the mechanism. □

References